

AI: More About Neural Networks and an Introduction to PyTorch

The previous article in this series on AI and machine learning was critical in many respects. First, we started using scikit-learn, a very powerful Python library for developing AI and machine learning based applications. Second, we deployed our first trained model. But, most important, the article introduced ChatGPT, maybe for the first time by a print magazine in India. This eighth article in the AI series talks a little more about neural networks as well as unsupervised learning, and introduces PyTorch, a very powerful framework capable of learning from data.



As mentioned in the last article, ChatGPT is a chatbot launched by OpenAI in November 2022. Within just two months of its existence, ChatGPT has created two opposing factions. One faction is supporting AI based tools like ChatGPT and waiting for a Utopian world created by AI. The other faction is worried about AI and making doomsday predictions. They fear a dystopian future — a *Terminator*-like scenario in which machines control

humans and not the other way round. Some of the people in this faction also believe that ChatGPT will be a job killer. The jobs potentially under jeopardy according to these people include business transcription, medical transcription, copywriting, teaching, legal work, etc. But do remember that half a century ago many people believed that computers were job killers and see how mistaken they were. Some people also predict that tools like ChatGPT might be the beginning of the

end for conventional search engines like Google, Bing, Baidu, etc — a belief I don't share personally.

But what about the popularity of ChatGPT? FaceBook took 10 months to get one million subscribers in 2004. Twitter took about two years to obtain one million subscribers in 2006. Instagram was downloaded by more than one million users within 2.5 months of it being made public in 2010. All are amazing achievements. But do you know how much time ChatGPT took to obtain one million subscribers after it was made available to the public? Less than a week! Simply amazing, isn't it? I don't know the exact number of subscribers as of writing this article. But nowadays the services of ChatGPT are often unavailable due to the large number of users.

What about DALL-E 2, also provided by OpenAI, a tool capable of generating images from descriptions. Will normal people ever be able to distinguish between images generated by master painters and the ones generated by tools like DALL-E 2. For the sake of art, I hope at least expert curators will be able to distinguish between the two, just like a gemologist can easily distinguish between a real and a duplicate diamond, a task not easy for common people.